Problem 5 - Machine Learning Solution

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The problem and the target

Warn maintenance engineers which components are likely to be pulled off an aircraft soon

2%
of records are removals (from the exam brief)

5 per 100
alerts at most, per 100 active components, each month

80%+
of actual removals detected, at least

- A component is an installed aircraft part, such as a hydraulic pump. An unscheduled removal is a component taken off without a plan; we predict it for the next 30 flight cycles (one cycle = one takeoff and landing; about a month of flying here)
- An alert is a component flagged for inspection. Each month every component gets a risk score and the top 5% are flagged
- Catching 80% of removals with only 5% flagged means about 1 alert in 3 is a real removal, so many false alarms are expected

Mock dataset (synthetic)

- No real data was shared, so we built a simulator
- 10,000 records: one per component every 30 flight cycles (shown as months): 80 aircraft x 5 components x 25 months; 2% are removals
- Each record: usage, 4 sensor readings, fault messages, maintenance history
- A hidden wear-out process makes sensors drift before a removal
- Label: 1 if the component is removed unplanned within 30 flight cycles

What we vary	Easy	Realistic	Hard
Removals with no warning signs	5%	15%	35%
Strength of warning signs	strong	medium	weak
Missing sensor readings	2%	6%	15%
Mislabelled removals	0%	1.5%	4%
Fleet changes in later months	none	mild	strong

Three difficulty levels, all with about 2% removals. Realistic is the headline.

Dataset used below: Realistic difficulty, one fixed random draw (seed 42), file data/mock_components.csv

Dataset schema

One row = one component at one monthly snapshot; features look back, the label looks 30 cycles ahead

Group	Key columns	Values
Identity and time	component_id, aircraft_id, snapshot_date	Serial number; 80 aircraft; 25 monthly snapshots
Component and fleet	component_type, aircraft_type, operator, part_number	5 component types (hydraulic pump, fuel pump, starter-generator, bleed-air valve, brake assembly); A320, B737, ATR72; airlines OP_A, OP_B, OP_C
Usage	flight_cycles_since_new, flight_hours_since_new, cycles_since_overhaul	How much the component has flown
Sensors (can be missing)	vibration, oil temperature, pressure, current: mean and std over the last 30 cycles	8 columns
Faults and maintenance	fault_count, last_fault_code, cycles_since_last_fault, days_since_last_maintenance	Fault messages and maintenance history
Label	removal_within_30_cycles	1 = removed without a plan within the next 30 cycles; about 2% are 1

29 columns in total (28 inputs and the label); the full list with units is in data/SCHEMA.md.

Full schema with units: data/SCHEMA.md. Data: data/mock_components.csv (10,000 rows, 29 columns)

Data preparation and splitting

Split by time: learn from the past, test on the future

Train	Validation	Test
60% of records months 1-15 (Jan 2022 - Feb 2023)	20% of records months 16-20 (Mar - Jul 2023)	20% of records months 21-25 (Aug - Dec 2023)

Missing readings
Marked as missing (it is itself a warning sign); gaps filled with a typical value where needed

Extreme values
Kept (they are often real failures); limited to a ceiling for the simple model

Categories
Component type, airline and similar labels are turned into numbers the models can read

Few removals
Only 2% are removals, so removals get extra weight in training and we judge by removals caught, not accuracy

About 200 removals in total. No leakage: records use only earlier months of the same component; test months are never used to make choices; a check on unseen aircraft gives recall 0.91-0.97 (boosting).

Data: Realistic difficulty, one fixed random draw (seed 42)

How the model decides

Engineered feature (x4 sensors)	Count	How it is computed
oil_temp_c_z_vs_type	4	(this month's reading - average of the same component type) / spread of that type
oil_temp_c_spread_z_vs_type	4	Same, for the within-month spread (std) of the reading
oil_temp_c_z_change_vs_previous	4	Deviation now minus deviation last month
oil_temp_c_z_slope_last_3	4	(Deviation now - deviation 2 months ago) / 2
oil_temp_c_z_vs_own_prior_mean	4	Deviation now minus this component's own average of its last 3 months
fault_count_prior_3, high_risk_fault_recent, history_length, oil_temp_c_is_missing	7	Faults in the 3 months before, serious code (F415/F520) in last 3 months, months of history, sensor missing flag
Copied as they are	23	Age, cycles, hours, faults, maintenance, raw sensor mean/std (18), plus 5 labels (component type, part number, aircraft type, airline, last fault code)

- 50 features in total. Averages and spreads come from the training months only; history uses earlier months of the same component, never the future
- The strongest are the deviation-from-type features; change since last month turned out weak
- A new component with no history gets missing history features and a low-confidence flag

The two models compared

	Logistic regression	Gradient boosting (chosen)
How it works	Adds up weighted clues into one score	Many small decision trees, each fixing the last one's mistakes
Strength	Simple, stable, easy to explain	Finds complex patterns and combinations of clues
Weakness	Misses complex patterns	Harder to explain, so we add plain-language reasons for each alert
Missing and extreme values	Filled with a typical value; extremes capped	Handled directly; extremes kept as they are

Both models get the same clues and the same split. They are compared over 20 seeds after the metrics slide, and the chosen model is reported on the slide after that.

Evaluation metrics and alert threshold

Metric	What it measures	How we use it
Recall@5% alerts (primary)	Share of true removals that fall in the top 5% highest-risk components of their month	Target ≥ 0.80, from the brief
Precision@5% alerts	Share of alerts that were real removals	Cost of false alarms for inspectors
PR-AUC	Area under the precision-recall curve: ranking quality over all cut-offs (random = base rate, about 0.02)	Compares models; suited to rare positives
95% CI and 20 seeds	Bootstrap over components (1,000 resamples); repeat on 20 independent seeds	Only 34 test removals, so show uncertainty

- Accuracy is not used: always predicting "no removal" is 98% accurate and catches nothing
- Alert rule (all results): top 5% by risk score each month = 20 of 400 components; model chosen on validation months, test months used once
- Fixed cutoff (drift signal): score 0.035, the top-5% level on the validation months, held fixed. On the test months it gives 3.95 alerts per 100 and the same 88% recall; a shift in alerts per 100 signals drift

Data: Realistic difficulty, seed 42; budget = 5% of active components per monthly period

Which model is better?

20 seeds	Logistic regression	Gradient boosting (chosen)	Reading
Recall@5%, mean ± std	0.74 ± 0.09	0.84 ± 0.08	Boosting about +0.10
95% CI of the mean recall	0.70-0.78	0.81-0.88	Intervals do not overlap
Seeds with recall ≥ 0.80	5/20	16/20	Target met far more often
Recall: higher model per seed (paired)	0 seeds	18 seeds (2 tied)	Wilcoxon $p < 0.001$
PR-AUC, mean ± std	0.43 ± 0.08	0.70 ± 0.07	Boosting ranks removals better
PR-AUC: higher model per seed (paired)	0 seeds	20 seeds (0 tied)	Wilcoxon $p < 0.001$

- Gradient boosting is better on both recall@5% (about +0.10) and ranking quality, and wins in most seeds, so it is the chosen model
- Logistic regression usually misses the 0.80 target (5 of 20 seeds); boosting meets it in 16 of 20, but its worst seed is 0.61: usually met, not guaranteed
- An untrained rule (sum of sensor-spread deviations) also scores about 0.74, so linear clues plateau there and trees add about 0.10 by combining clues
- Both models are evaluated identically; the std of about 0.08 per seed is why a single split such as seed 42 cannot rank them

Data: Realistic difficulty, 20 independent seeds (1-20); each repeats the whole pipeline

Results of the chosen model (20 seeds)

Test set (per seed)
400 components x 5 months = 2,000 component-months; removals per seed: mean 39 (29-51)

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Alert budget
Top 5% per month = 20 of 400 components; over 5 months = 100 alerts per seed

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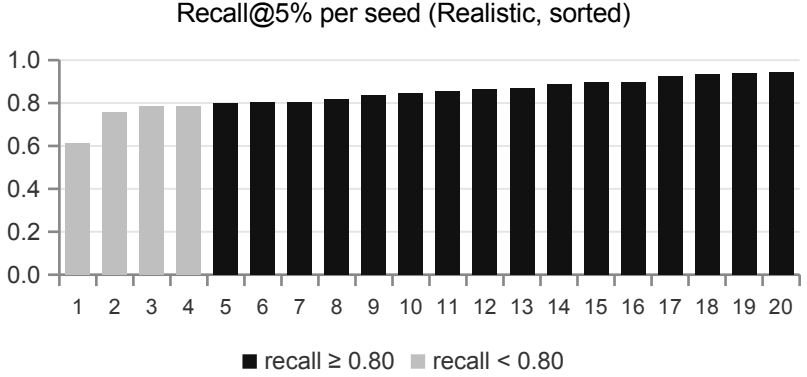
Outcome (mean over seeds)
32 real removals found, 68 false alarms, 6 removals missed

0.84 ± 0.08
recall@5%, mean ± std (95% CI of the mean 0.81-0.88)

0.32 ± 0.05
precision: share of the 100 alerts that were real removals

0.70 ± 0.07
PR-AUC: ranking quality over all cut-offs; random is about 0.02

16 of 20
seeds with recall at or above the 0.80 target

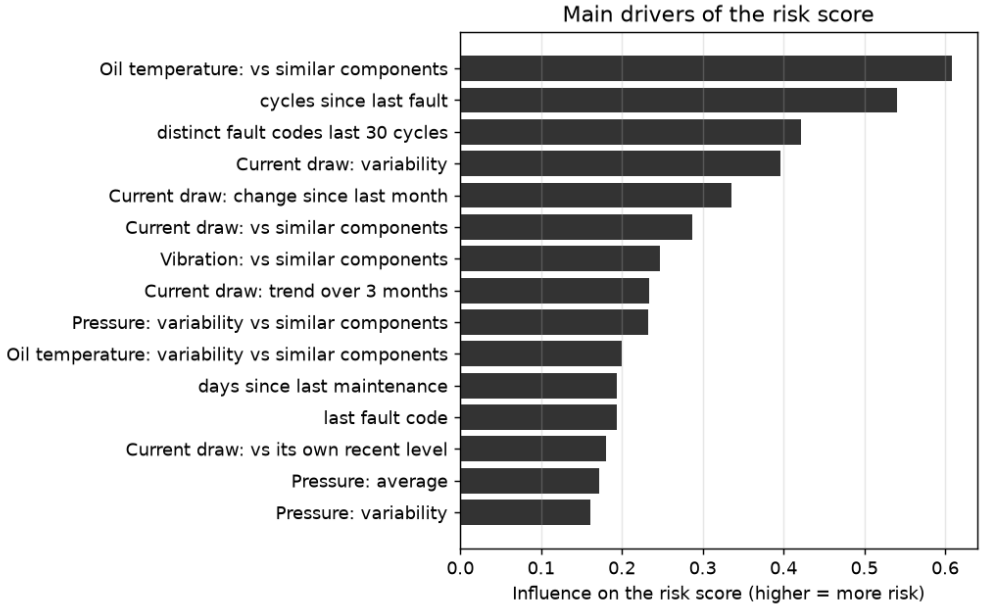


Profile	Recall@5%, mean ± std	Min-max	Seeds ≥ 0.80
Easy	0.94 ± 0.04	0.84-1.00	20/20
Realistic	0.84 ± 0.08	0.61-0.94	16/20
Hard	0.66 ± 0.08	0.50-0.80	1/20

Shipped dataset (seed 42): recall 0.88 (30 of 34 removals), bootstrap 95% CI 0.76-0.97. The target is usually met, not guaranteed: the Realistic worst seed is 0.61. Hard: 35% of removals give no sensor warning.

Data: Realistic difficulty, 20 independent seeds (1-20); Easy and Hard rows use the same method

What drives the risk score (SHAP)



- SHAP shows how much each of the 50 features pushed one component's score up or down
- Input: that component's features for the month, scored by the boosting model
- Bars: average size of the push over the validation months
- Top drivers: oil temperature vs similar components, cycles since last fault, distinct fault codes, current draw
- Reasons show what the model relied on, not confirmed causes

Data: Realistic difficulty, seed 42; SHAP values on validation months, boosting model

Three alerts explained

Alert = a component in the top 5% by risk score that month (20 of 400). Outcome is judged over the next 30 flight cycles.

Case	Component, month	Risk score (0-1) and rank of 400	Top reasons (SHAP)	Outcome
True positive: flagged and removed	Hydraulic pump C00542, Sep 2023	0.99 (1st)	Pressure readings unstable (4.1 std below normal); oil temperature 1.7 std above similar components; 16 days since last maintenance	Removed within 30 cycles: correct alert
False positive: flagged, not removed in time	Bleed-air valve C00592, Nov 2023	0.88 (2nd)	Pressure unstable; oil temperature 3.4 std above similar components; current draw rising; 5 faults in the last 30 cycles	Removed about 1 month later: counted as a false alarm, but the warning came first
False negative: removed, not flagged	Hydraulic pump C00471, Dec 2023	0.003 (229th)	No strong signal: largest sensor deviation only 1.2 std; current draw slightly up; 79 days since last maintenance	Removed with no warning: a sudden failure no model can see

Risk score = predicted chance of removal within 30 cycles. Rank = position among the 400 components scored that month. C00542 etc. are component IDs (serial numbers). Top reasons = the features with the largest positive SHAP values for that component and month (SHAP: how much each feature pushed the score up, slide 11), in plain words.

Data: Realistic difficulty, seed 42, test months. std = standard deviations from the average of similar components of the same type

Cold start, missing data, drift

Situation	How the model handles it	What we tested (recall@5%)	How we monitor it
New components (cold start)	Clues compare a component with its type average, so it still gets a score from current readings; history features are missing and the row gets a low-confidence flag	New components fail early: 0.90. One type never seen: 0.65. All types never seen: 0.18	Share of low-confidence components; new types: build baselines, retrain
Missing data	Missing sensor readings stay missing, with an is-missing flag per sensor (a stopped sensor is itself a warning sign); boosting handles gaps natively, logistic uses the median from training	20% more missing readings: 0.76 (was 0.88)	Missing rate per sensor; a jump means a broken data feed: fix the feed
Model drift	Fixed training baselines; stress-tested on simulated shifts; retrain on a trigger, the new model must beat the current one and a person approves	Sensor offset or 30% more flying: 0.88. Failure rate doubled: 0.63	PSI/KS on inputs per airline; alerts per 100 at the fixed cutoff; rolling recall below 60% for 3 months

PSI (Population Stability Index): compares the share of rows per value range now vs in training; below 0.1 stable, above 0.25 major shift.
KS (Kolmogorov-Smirnov): the largest gap between the two cumulative distributions, from 0 (same) to 1. Both need no removal labels.

All results are on our own simulator, so robustness is shown only for the shifts we modelled.

Data: Realistic difficulty, seed 42; recall@5% on test months re-generated with one change (next slide)

Robustness to data drift

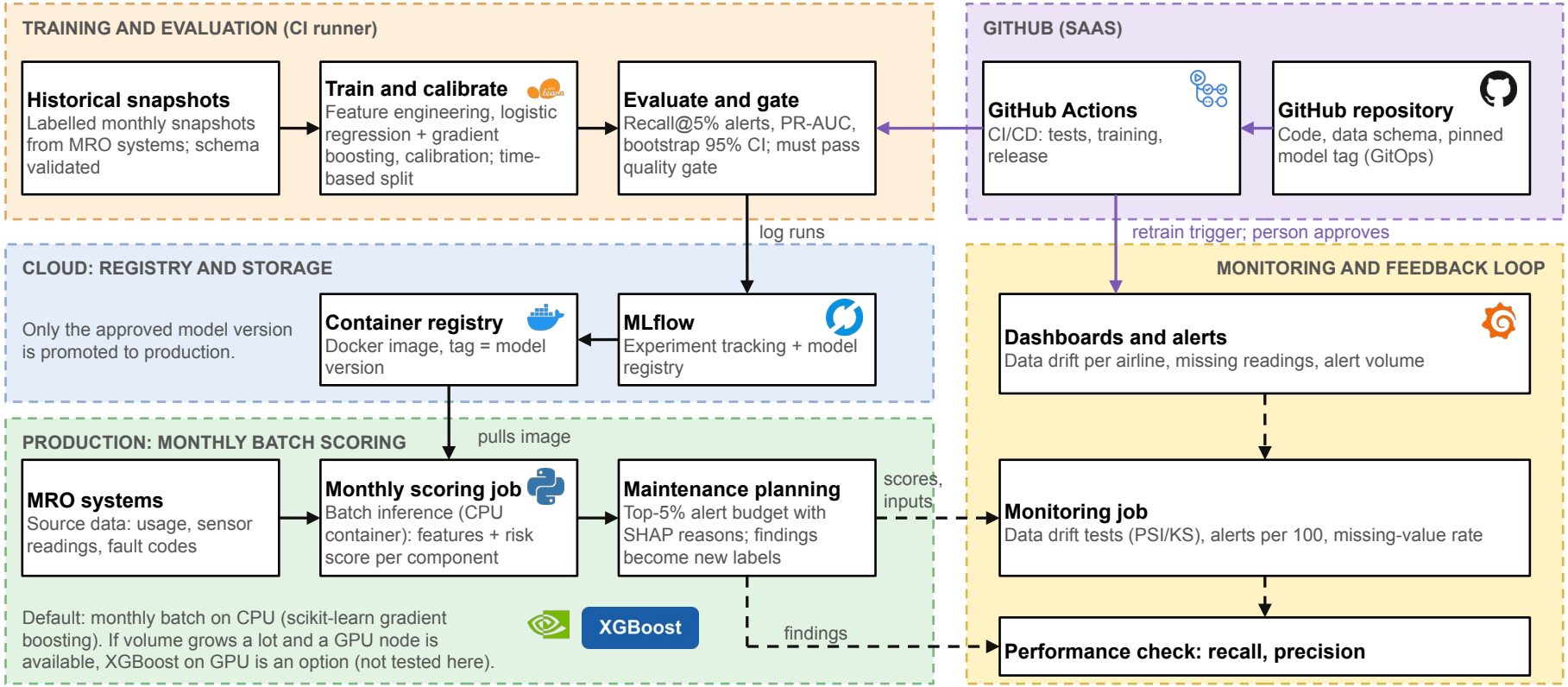
Simulated drift: test months re-generated with one change, scored by the already trained model.

What changed	Drift type	Removals caught	Alerts per 100 at the fixed cutoff	How we would detect it
Nothing (baseline)	-	88%	4.0	-
One airline's oil sensors read high	Covariate shift	88%	4.0	PSI/KS per airline
30% more flying	Covariate shift	88%	4.0	PSI/KS on flight hours
20% more missing readings	Data-quality shift	76%	4.9	Missing rate per sensor
Failures twice as frequent (78 removals)	Prior (base-rate) shift	63%	8.3	Alerts per 100 jump
New components fail early	Concept drift	90%	4.6	Rolling recall; low-confidence flag
One component type never seen	Unseen category	65%	3.0	Low-confidence flag
All component types never seen	Unseen category	18%	0.1	Alerts per 100 collapse

- Robust to covariate shift; weak on unseen component types and when the failure rate doubles
- Normal rule: always 5 per 100. Fixed cutoff (score 0.035, set on validation): alert volume moves with the data, so it signals drift

Data: Realistic difficulty, seed 42; test months regenerated with one change; model trained on the original data

Production architecture (proposed)



Adapted from an MLOps reference architecture

Data, artifacts — Control (GitHub) — Telemetry, findings

Monitoring and retraining

What we watch	How	What it catches	Action
Inputs per airline	Compare with training data (PSI and KS: scores for how much data changed)	One airline's sensors drifting; hidden when the fleet is pooled	Check sensors and data feed
Missing readings	Share missing per sensor	A broken data feed	Fix the feed
Alerts per 100	Count under a fixed score cutoff	A change in the failure rate	Review capacity if it moves by more than 2
Low-confidence share	Share of components marked low-confidence	New component types	Build baselines, retrain
Recall and precision	Past alerts vs removals after 30 cycles	The model getting worse	Retrain if recall < 60% for 3 months

- These numbers are starting points: even our unshifted test months show a high PSI on flight hours, so thresholds are tuned on normal month-to-month variation
- Retrain every quarter or on a trigger: time split, new model must beat the current one, a person approves

Limitations and future work

Limitation	Planned next step
Simulated, easy data: an untrained rule already reaches recall 0.74	Validate on real maintenance data under a data-sharing agreement
Only 34 removals in the test months: a single recall estimate can be off by about 10 points	Re-run the evaluation on more months as they arrive (rolling backtests), keeping confidence intervals
Weak spots: unseen component types (recall 0.65) and a doubled failure rate (0.63)	Handle new component types better; automatic drift reports and safe model updates
No cost of an alert vs a missed failure; '30 cycles = 1 month' is a simplification	Set the alert threshold from real inspection and disruption costs; predict time to removal instead of a fixed 30-cycle window
No fault chapter codes, flight phase, operating conditions or 'no fault found' labels	Add these features and labels to the schema